

[illegible]

Network: Polygon PoS Mainnet (Chain ID: 137)

Report ID: rep_matic_advanced_090 | Tier: ADVANCED | Application Surface: SHIELD

Created At: 2026-09-10T12:36:16.909Z

ELEVATED RISK (UNVERIFIED CODE) (Risk: 72/100 | Quality: 96/100)

Release Decision: **BLOCKED** | Verification Status: **INSUFFICIENT EVIDENCE**

STOP-SELL DIRECTIVE: INSUFFICIENT EVIDENCE: Coverage below minimum threshold for institutional sale

Confidence Score:

60/100

Evidence Coverage:

20%

[VERIFIED - IMMUTABLE]

Root: sha256:fe09a03d655c24ac...a945dee20d5b

Standard: Merkle Non-Repudiation Seal | RFC 3161 Pinned | Verification: </api/audit/report-pdf>

L1 Network Resilience Matrix: Consensus Verification 0% | P2P Topology Graph 0% | Node Fault Detectors 100% | Cryptographic Verification 0%

L1 Network Manifest: Network Polygon PoS Mainnet | Consensus: Unverified Source | Client: Polygon Proof of Stake | Cryptography: secp256k1 / BLS

Contract lacks verified on-chain runtime bytecode. In accordance with Velmère evidence standards, missing bytecode results in an elevated risk assessment (72/100).

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Summary of distributed architecture and consensus mechanism

Protocol / Asset: Polygon Proof of Stake

Architecture Type: Native Base Layer (Layer-1 - No Smart Contract)

Network Identifier: native chain:matic

Native Symbol: MATIC

Rule Governance: Distributed node consensus across validators / miners

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Analysis of block validation rules, transactions, and cryptographic curves

Verification of Polygon Proof of Stake is conducted against native Layer-1 consensus specifications. EVM bytecode decompilation is NOT APPLICABLE for native base protocol assets.

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Identified risk vectors in base layer architecture

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Decentralization evaluation, validator distribution, and improvement proposal process

* Network Consensus Governance	Distributed Node Consensus	VERIFIED
* Smart Contract Admin Keys	NOT_APPLICABLE (Native Layer-1 Protocol)	VERIFIED
* Emergency Network Pause Capability	NOT_APPLICABLE (Decentralized Network)	VERIFIED
* Arbitrary User Balance Drain	NOT_APPLICABLE (Cryptographic Keys Enforced)	VERIFIED

Protocol rules for Polygon Proof of Stake are enforced by distributed node consensus; no privileged smart contract owner key exists (NOT APPLICABLE).

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Global market liquidity, supply velocity, and mempool dynamics

* Market Reserve Liquidity	Global Spot Market Liquidity	VERIFIED
* DEX AMM Pairs	NOT APPLICABLE (Native Base Asset)	NEUTRAL
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* Transfer Tax / Sell Fee	NOT APPLICABLE (Standard Network Fee Model)	VERIFIED

VELMÈRE ADVANCED AUDIT REPORT

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Liquidity for Polygon Proof of Stake is anchored in global spot venues and custodial settlement networks; automated market maker liquidity locks are NOT APPLICABLE.

NETWORK ATTACK VECTORS & CONSENSUS RESILIENCE (PRO)

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Modeling of chain reorganizations, Sybil attacks, and P2P partitioning

* External Oracle Dependency	NOT_APPLICABLE (Protocol Native Settlement)	VERIFIED
* Flash Loan Attack Vector	NOT_APPLICABLE (Non-EVM Execution Environment)	VERIFIED
* Double-Spend Attack Vector	Secured by network consensus	VERIFIED

STATE ENGINE DETERMINISM & MULTI-CLIENT WIRE COMPATIBILITY (ADVANCED)

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Binary compatibility verification and state transition engine determinism

* State Transition Determinism	Conforms to protocol specification	VERIFIED
* Client Implementation Diversity	Reference implementations evaluated	VERIFIED

State integrity for Polygon Proof of Stake is enforced by deterministic state transitions; EVM storage layout diffing is NOT APPLICABLE.

PROTOCOL HARD/SOFT FORK EVOLUTION & INVARIANT PROOFS (ADVANCED)

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Tracking of consensus rule evolution and backward compatibility guarantees

Protocol Status: Active Production Mainnet

Consensus Fuzzing: NOT EXECUTED (Engine not commissioned)

HUMAN ANALYST REVIEW & PROTOCOL ARCHITECTURE AUDIT (ADVANCED)

REVIEW STATUS: AUTOMATED ANALYSIS ONLY (AUTOMATED_ONLY)

Attestation by independent distributed systems auditor

Review status: INDEPENDENT HUMAN REVIEW NOT COMMISSIONED. In accordance with Velmère policy, human review claims are NEVER emitted unless verified reviewer evidence exists.

Reviewer State: INDEPENDENT HUMAN REVIEW NOT COMMISSIONED

Analyst: NONE (Independent manual review not commissioned)

Attestation Signature: NONE (No attestation emitted - review not performed)

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